CURRICULUM VITÆ

Wenyu Chiou

Ph.D. Candidate · Civil & Environmental Engineering | Complex Adaptive Water Systems (CAWS) Lab, Lehigh University

I study human decision-making under flood risk and evaluate whether large language models (LLMs) reproduce empirically observed response distributions and decision-making relationships. My research combines survey analysis, behavioral simulation, and governed agents to assess human-water interactions and household financial outcomes.

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§ 01 Education

Ph.D., Civil & Environmental Engineering, Lehigh University — Aug 2024 – Expected December 2027
Complex Adaptive Water Systems (CAWS) Lab · Advisor: Prof. Y.C. Ethan Yang · Center for Catastrophe Modeling and Resilience (CMR)
Dissertation: *Advancing Coupled Human-Flood Modeling: From Empirical Foundations to LLM-Driven Multi-Agent Simulation*
Research: human-grounded LLM evaluation; agent-based modeling (ABM) coupled with catastrophe simulation; Water Agent Governance Framework (WAGF) for constrained decisions in human-water simulations.

M.S., Hydrological and Oceanic Sciences, National Central University (NCU), Taiwan — Sep 2021 – Jun 2023
Surface Hydrology Group · Institute of Hydrological and Oceanic Sciences (IHOS) · GPA 4.3/4.3 · Dean's Award, College of Earth Sciences (Jun 2023). Thesis: groundwater flow characteristics of the Taoyuan coastal zone.

B.S., Earth Sciences, National Central University (NCU), Taiwan — Sep 2017 – Jun 2021
GPA 3.9/4.3 · Best Undergraduate Student Project and Report (Nov 2020).

§ 02 Research Experience

Graduate Research Assistant — Complex Adaptive Water Systems (CAWS) Lab, Lehigh University — Aug 2024 – present

- Co-authored the *Water Resources Research* (2026) article substituting large language models as the behavioral engine of a human–water simulation — published; second author of two.
- **Human-grounded LLM evaluation:** Built a Python/Ollama workflow comparing survey response distributions and decision-making pathways with Gemma 4:12B. Ten-run median structural equation modeling (SEM) estimates matched human associations in direction and significance on 5 of 12 pathways; social-capital response standard deviation was 74-76% lower for this instrument.
- **WAGF:** Designed a governance layer for human-water simulations checking physical, financial, and behavioral constraints plus rating-reason-action consistency. Targeted retry feedback and audit traces support correction of invalid proposals before state updates. Research software in preparation for release.
- **FLOODABM:** Coupled household-agent and catastrophe simulations with National Flood Insurance Program (NFIP) rules to assess damage, payouts, and out-of-pocket exposure. In 2011-2023 model scenarios, adaptation reduced homeowner physical damage by 5.8% and losses after insurance by 45.9% versus no adaptation, showing why damage alone understates financial protection. Manuscript under revision.

Research Assistant — Surface Hydrology Group, IHOS, National Central University — Jan 2024 – Jun 2024

- 2D/3D groundwater numerical modeling and calibration for the Taoyuan coastal zone (HydroGeoChem); evaluated nature-based solutions for groundwater and watershed management.

Summer Research Intern — National Science and Technology Center for Disaster Reduction (NCDR), Taiwan — Summer 2023

- Processed downscaled climate-model ensembles for Taiwan for the Intergovernmental Panel on Climate Change Sixth Assessment Report (IPCC AR6); automated multi-model comparison pipelines in Python and MATLAB.

Undergraduate Research Intern — Institute of Earth Sciences, Academia Sinica — Jul – Aug 2020

- Stable-isotope hydrology ($\delta D/\delta^{18}O$), Feitsui Reservoir.

§ 03 Publications & Presentations

PEER-REVIEWED

Yang, Y. C. E., & **Chiou, W.** (2026). Leveraging large language models for agent-based simulation of human-water system interactions. *Water Resources Research, 62*(6), Article e2025WR042111. https://doi.org/10.1029/2025WR042111

MANUSCRIPTS & SUBMISSIONS

Chiou, W., Yang, Y. C. E., Tanaka, T., Jamrussri, S., & Feng, S. (2026). *Household flood adaptation and financial outcomes: A coupled human-flood modeling analysis of homeowners and renters* [Manuscript under revision]. *Water Resources Research*.

Chiou, W., & Yang, Y. C. E. (2026). *Flood adaptation decision variables and pathways across social groups: A comparison between human survey responses and responses generated by large language models* [Manuscript in preparation]. Planned submission: *Progress in Disaster Science*.

CONFERENCE PRESENTATIONS

Chiou, W., & Chiou, H. (2026). *What makes AI believable? Comparing AI and human responses in mental health risk assessment* [Recorded presentation]. International Society for Data Science and Analytics (ISDSA) Annual Meeting.

Chiou, W., Yang, Y. C. E., Tanaka, T., Jamrussri, S., & Feng, S. (2025). *Modeling long-term household flood adaptation under social heterogeneity: A coupled agent-based modeling framework* [Poster presentation, NH41E-0449]. American Geophysical Union (AGU) Fall Meeting.

Chiou, W., Yang, E., Sapat, A., Casagrande, D., Tanaka, T., Otsuyama, K., & Yotsui, S. (2025, July 21). *Investigating the decision-making process and causal relationships between flood risk perception and adaptation actions* [Oral presentation, 38-03]. International Sociohydrology Conference (ISHC), Tokyo, Japan. Conference program

Additional conference activity: AGU Fall Meeting (2023), submarine groundwater discharge research using electrical resistivity tomography, field observations, and numerical simulation (conference abstract).

§ 04 Awards & Fellowships

Dean’s Award, College of Earth Sciences — National Central University — Jun 2023

Interdisciplinary Research Fellowship (Hydrology/Geophysics/Geology) — NSTC, Taiwan · $4,000 USD/year — 2021 – 2023

Best Undergraduate Student Project and Report in Earth Sciences — National Central University — Nov 2020

Undergraduate Student Research Project Grant — NSTC, Taiwan · $1,600 USD total — 2020 – 2021

§ 05 Research Interests

- Human-grounded LLM evaluation
- Behavioral and social simulation
- Computational social science
- LLM multi-agent systems
- Agent evaluation & governance
- Agent-based modeling
- Catastrophe modeling
- Human–water interaction
- Decision-making under risk
- Structural equation modeling

§ 06 Methods & Tools

QUANTITATIVE METHODS

Statistical modeling · psychometrics · confirmatory factor analysis (CFA) · SEM · Bayesian calibration · Monte Carlo · sensitivity and counterfactual analysis · experimental design

AGENT SYSTEMS

LLM evaluation · structured generation · prompt/context engineering · multi-agent workflows · validation rules · audit logs · simulation coupling

PROGRAMMING

Python · TypeScript · R · MATLAB · C++

SOFTWARE & TOOLS

Git · pytest · GitHub Actions / continuous integration (CI) · PyPI · command-line and REST API development · reproducible pipelines · Jupyter · SPSS Amos · ArcGIS Pro · HydroGeoChem

AI TOOLING

Ollama · LangChain · OpenAI/Anthropic APIs · Model Context Protocol (MCP) · Codebase Memory MCP · Claude Code · Codex CLI · Cursor · plugin/skill development

§ 07 Research Software · LLM Agent Systems & AI Research Workflows

research-hub

AI-operable research workspace · PyPI v1.1.1 · Maintained open-source tool

An AI-operable research workspace for Zotero, Obsidian, and NotebookLM — search, ingest, and sync papers through one CLI, MCP server, and REST API. Listed in Awesome MCP Servers.

FLOODABM

Coupled ABM × CAT model · Research prototype, archived

A large-scale simulation (52,141 households) that quantifies human–flood interaction and insurance outcomes; Zenodo-archived companion code with CITATION.cff.

codex-delegate

AI research workflow tooling · Maintained open-source tool

Agent-delegation wrappers that verify change from git state, not the delegate's own report; a regression test pins a real upstream failure. CI on Ubuntu and Windows.

fable-method-harness

Portable AI-agent operating harness · Maintained open-source tool

Cost/model routing with an audit trail for AI-assisted research and engineering workflows.

WAGF — Water Agent Governance Framework

LLM-agent simulation governance · In preparation for release

A governed LLM-agent framework for human–water simulation that checks agent decisions against physical, financial, and behavioral-theory constraints before simulation-state updates.

agent-collab-skills

Multi-agent orchestration · Maintained open-source tool

Multi-agent orchestration — task splitting, output reconciliation, adversarial debate, shared memory, and a pre-merge acceptance gate.

ai-research-skills

Research workflow · Maintained open-source tool

15 reusable skills across eight research stages for literature collection and organization, gap identification, research-question formulation, study planning, and manuscript preparation; 250+ GitHub stars.

§ 08 Professional Service

Graduate Student Committee Member — Lehigh University, Dept. of Civil & Environmental Engineering 2025 – present

§ 09 Community & Open-Source

awesome-agentic-ai-zh 7.0K+ GitHub stars and 946 forks (checked September 2026)

Trilingual (zh-TW / zh-Hans / EN) 7-stage roadmap from LLM basics to production multi-agent systems; automated checks keep all three languages in sync; external contributors.

Selected open-source engineering contributions

Root-caused an engine-selection bug and added a regression test in BuilderIO/agent-native; fixed Windows file-permission handling in langchain-ai/openwiki; and added first-class Taiwan-market and institutional-flow data support to daily_stock_analysis.

Languages Mandarin Chinese — native · English — professional